

Planning Report

SSY226: Design Project in Systems, Control, and Mechatronics

**Real-Time Obstacle Avoidance for Mobile Robots using Deep
Reinforcement Learning and Model Predictive Control
(MPC in the Loop)**

Project Group 14.2

Mathanesh Vellingiri Ramasamy

Rishikesh Vishnu Sivakumar

Shao-Hsuan Huang

Valdemar Samuelsson

Yu Kang



CHALMERS
UNIVERSITY OF TECHNOLOGY

Submitted on Friday 11th September, 2026

Background

An efficient, safe, and collision-free navigation is crucial for **autonomous mobile robots (AMRs)** in an industrial environment for various applications like material handling, logistics, etc. The industrial transition towards to use of AMRs is increasing due to its improvement in efficiency and productivity. However, there are certain challenges when it comes to real-time obstacle avoidance in a challenging, dynamic environment which we need to address in-order to provide an optimal fleet of AMRs. This study could be helpful in real world autonomous driving systems requiring collision free navigation.

In the previous work [2], we can see that the integration of **Deep Reinforcement Learning (DRL)** along with **Model Predictive Control (MPC)** has significantly improved the obstacle avoidance performance. The integration was done in such a way that the DRL was used to generate reference trajectories in complex environments while MPC refined those trajectories based on the kinematic and dynamic constraints of the robot as well as the current and predicted future positions. The purpose of the hybrid approach is to utilize the DRL's capability to navigate non-convex obstacles which are challenging for MPC, and utilize the MPC to optimize the trajectories produced by the DRL network. This hybrid approach allows us to achieve a better computational efficiency, time to destination, and adaptability to complex, dynamic and multi-robot situations when compared to the pure MPC or pure DRL approaches. It can be seen that the proposed hybrid model (HYB-DDPG) achieved a success rate of 100% in all tested scenarios, significantly outperforming the MPC and DQN-based algorithms.

As a follow up project, here we focus on exploring ways to improve the existing hybrid model (HYB-DDPG) by optimizing the trajectory generation from the DRL model or other hybrid methods in-order to obtain a better computational time and time to destination.

Purpose

The purpose of this project is to enhance the performance of the current DRL-based trajectory planner by incorporating MPC in two ways.

- Incorporating MPC into the DLR-learning phase to improve the neural networks performance.
- Utilizing DRL in the MPC algorithm to improve its performance.

By exploring these approaches the goal is to develop a hybrid DLR-MPC model that outperforms the current trajectory planner. The aim is to create a unified system without the need to switch between DRL and MPC, providing consistent behavior across all time steps and improving performance.

Problem Statement

Based on previous work, the purpose of this project is to explore a better method for combining MPC with DRL to improve the efficiency of computation and control effectiveness, and also try to find a good alternative for the current switch method between A* algorithm and DRL method.

In this task, we will focus on the following tasks. Firstly, in order to more effectively balance short-term and long-term benefits, the Q-learning function will be used to replace the final cost of MPC, in order to better evaluate the long-term effects in the future. During this process, the Q-learning function is further trained and updated after each MPC control operation. The structure of the MPC controller needs to be re-designed to incorporate the Q-learning function.

Secondly, in order to optimize the computation time of the MPC controller, it will be added to the reward of deep reinforcement learning. We will use an approximation function, which take the computation time into consideration, to replace the Q-function and retrain it.

Finally, we need to verify whether the updated control model is stable and compare the improvement in the model performance.

Boundaries

This project focuses solely on improving the existing DRL with MPC framework through simulation studies. Hardware implementation and physical testing on real robots are outside the project scope due to time and resource constraints. In addition to this, detailed theoretical analysis and mathematical proofs are not included. These boundaries are established to maintain focus on enhancing the simulation-based framework while leaving more resource intensive aspects for subsequent works.

Methodology

We have three methods proposed to be implemented to have a better integration of DRL-MPC hybrid model. We list them as below:

- Firstly, since the computation of the MPC can not be guaranteed to be converged or solvable in acceptable computational time for like dynamic objects or non-convex obstacles, we can instead modify the cost function of the MPC to consider the Q-function of the DRL model. In other words, we calculate the optimal path by MPC considering the dynamics of the moving objects and also the reward values of Q function. Since now we are considering the Q function straightforwardly in MPC cost function to calculate optimal paths, it is expected to be easier to generate a reasonable path, comparing to generate reference path switching between A* algorithm and the DRL method then fed into MPC to generate paths, which is the

current method and is considered to be more complex. The computational time is expected to be shortened and we might still be able to retain the success rate of the HYB-DDPG model.

- Secondly, to deal with the computational time in a different way, we can, on the other hand, focus on fixing the trajectory given from the DRL model. That is, if we modify the action space definition to let MPC easier to solve, we might be able to compute the optimal path in a shorter time. For example, we can change the action space to the horizon for optimization of MPC which might be easier to used by MPC that can save some time for transforming action space while calculating. Also, we can further consider the computational time of MPC into the DRL model's Q-function to generate a trajectory later calculated by MPC that is rewarded for shorter computational time, which might make the computational time even shorter. Since the overall hybrid model structure is not changed a lot, the performance might not be influence a lot but we can have a shorter time for computation.
- In previous methodology just mentioned above, we want to create an easier trajectory and action representation for MPC to compute easier. However, from TABLE I in [2], we know that the shortest computational time, no matter in which scenario, **DDPG (Deep Deterministic Policy Gradient)** model outperforms other methods a lot, the only problem if that the model does not gurantee to be stable or optimal. Therefore, if we can make it stable and as optimal as it could, we can have an acceptably optimal, stable and fast solution. To achieve this, we can consider the object function of MPC into Q-function to make sure optimality, and also the stability check and feasibility check in to make sure stability. Also, if we are concerning that the computational time might be longer since now the Q-function is complex, we can also consider the overall computational time in Q-function to try to achieve a quick computation. Summing up, we use the concept of MPC in Q-function and use the advantage of the computational efficiency of DRL (No further computation of MPC) to try to achieve a fast, stable and acceptably optimal solution.

These new models will be evaluated against the the current previous model using the existing simulation environment. The evaluation criteria will include computation time, finish time steps and success rate. The model will be iteratively based on the evaluation results, with the goal of outperforming the current model on at least on metric while maintaining or improving the others.

Time Plan

This is a rough estimation of how our workflow will look like for this project.

Week 47 18-Nov-24 - 22-Nov-24		Week 48 25-Nov-24 - 29-Nov-24		Week 49 02-Dec-24 - 06-Dec-24		Week 50 09-Dec-24 - 13-Dec-24	
Activities	List of activities	List of activities	List of activities	List of activities	List of activities	List of activities	List of activities
	Brainstorm session	Familiarize with previous works (including codes and materials)	Finish the prototype model	Refine the model based on validation			
	Literature Review	Experiment with the methodology (Extract Q-function)	Start the formal simulation test (Validation)	Obtain an initial acceptable test result			
	Familiarize with previous works (including codes and materials)	Initiate and proceed with a prototype model (plan to complete 40%)					
	DEADLINE [18-Nov-24]: Planning Report						
Week 51 16-Dec-24 - 20-Dec-24		Week 52 & Week 01 23-Dec-24 - 01-Jan-25		Week 02 02-Jan-25 - 07-Jan-25		Week 03 08-Jan-25 - 17-Jan-25	
Activities	List of activities	Christmas	List of activities	List of activities	List of activities	List of activities	List of activities
	Complete the model		Refine the draft report	Refine and submit the final report			
	Refine and fine-tune to get best result		Start with the final report	Submit the paper review			
	Start with the draft report				DEADLINE [10-Jan-25]: Paper Review		
					DEADLINE [17-Jan-25]: Paper Evaluation		
			DEADLINE [07-Jan-25]: Draft Report	DEADLINE [17-Jan-25]: Final Paper			

Figure 1: *Project Time Plan*

References

- [1] Collision-Free Trajectory Planning of Mobile Robots by Integrating Deep Reinforcement Learning and Model Predictive Control (2023) (CASE).
(Authors: Ze Zhang, Yao Cai, Kristian Ceder, Arvid Enliden, Ossian Eriksson, Soleil Kylander, Rajath Sridhara and Knut Åkesson)
- [2] Bird's-Eye-View Trajectory Planning of Multiple Robots using Continuous Deep Reinforcement Learning and Model Predictive Control (2024) (IROS).
(Authors: Kristian Ceder, Ze Zhang, Adam Burman, Ilya Kuangaliyev, Krister Mattsson, Gabriel Nyman, Arvid Petersén, Lukas Wisell and Knut Åkesson)
- [3] Temporal Difference Learning for Model Predictive Control.
(Authors: Nicklas Hansen, Xiaolong Wang, Hao Su)